

MATH 1342 – Elementary Statistics – Week 4: Percentiles, Boxplots and Two-Variable Relationships

Sep 15 – 17 | 2.3 One Quantitative Variable: Percentiles • 2.4 Two Variable Relationships | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the sample standard deviation of a small dataset by hand. (CLO 2) 2. Compute and interpret z-scores to compare values measured on different scales or in different groups. (CLO 2) 3. Find the five-number summary, range, and IQR; apply the $1.5 \times \text{IQR}$ rule to identify outliers; and construct or interpret a boxplot. (CLO 2)	e-Text: 2.3 One Quantitative Variable: Using Percentiles Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles Slides: 2.3 One Quantitative Variable: Using Percentiles	Participation: One Quantitative Variable (Kahoot!) – Tue: review of 2.2 Participation: Using Percentiles – Your Turn: comparing heights with z-scores Participation: Box Plots and Outliers – five-number summary, $1.5 \times \text{IQR}$ rule	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM)	WileyPLUS StatKey Lecture Slides (course website) Kahoot! Zoom (live class)
4. Compare a quantitative variable across groups with side-by-side boxplots, and compute and interpret a difference in means. (CLO 2) 5. Describe the direction, form, and strength of the relationship in a scatterplot and estimate the correlation r. (CLO 7) 6. Use technology to compute r and explain how outliers, nonlinear patterns, and the absence of causation limit its interpretation. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Quick Self-Quizzes: tea vs. coffee boxplots and difference in means; NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 20, 11:59 PM)	WileyPLUS Lecture Slides (course website) Zoom (live class)
<i>Items that cover objectives from more than one row above (objectives in parentheses)</i>			Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.4 (all objectives)	Assignment Tool in Blackboard

CLO 2 Recognize, examine, and interpret the basic principles of describing and presenting data. **CLO 7** Solve linear regression and correlation problems.